

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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STG327

Deliver performance at scale with Amazon S3

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Agenda

- 01 Performance philosophy
- 02 Using the scale of Amazon S3
- 03 Scaling dimensions
- 04 Implementing best practices
- 05 Key takeaways

Performance philosophy



Thinking about Amazon S3



Trillions of objects and
millions of requests per
second



Simple GET/PUT
interface to your
objects



Distributed service
with multiple
endpoints

Core tenets of performance

1

**Price tracks
performance**

2

Predictability

3

Scalability

Price tracks performance

Pay for performance based on your application's needs



S3 Standard



S3 Standard-IA



S3 One Zone-IA



S3 Glacier
Instant Retrieval



S3 Glacier
Flexible Retrieval

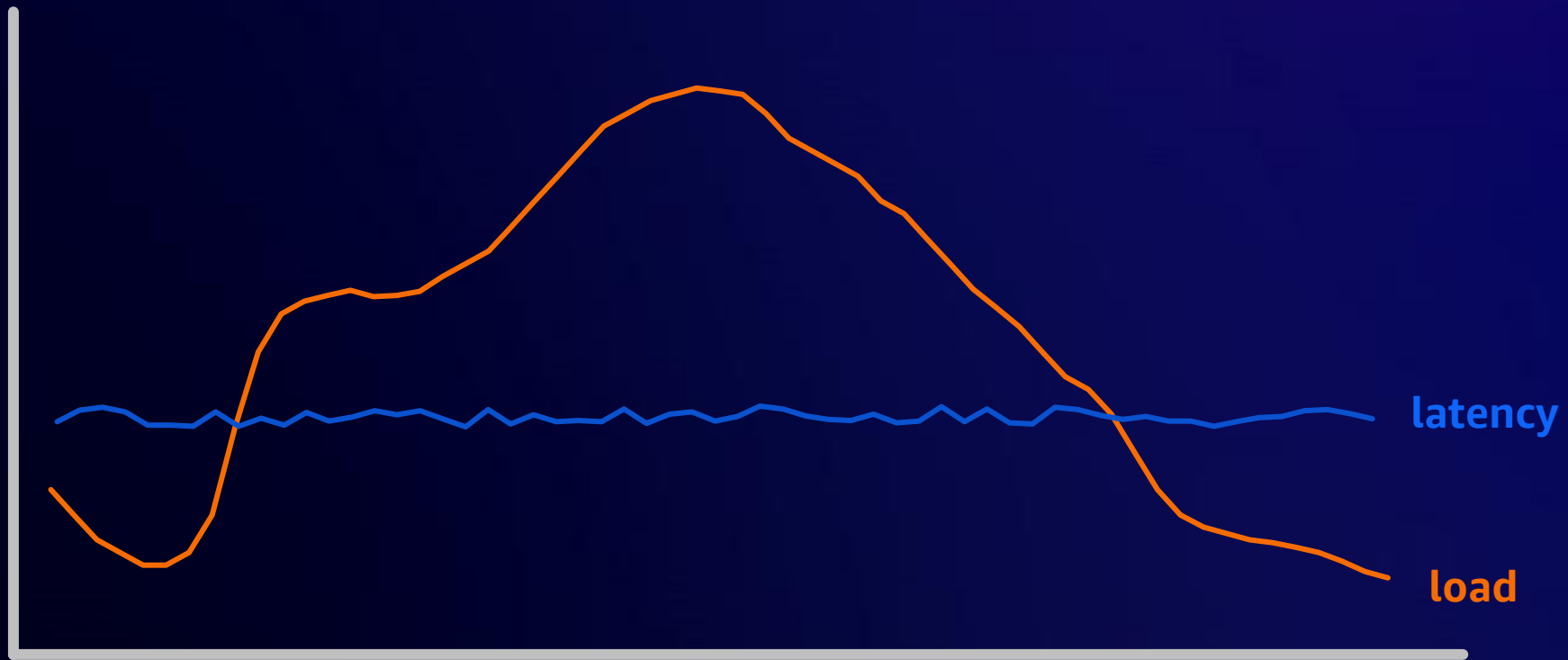


S3 Glacier
Deep Archive



Predictability

Amazon S3 latency doesn't vary much, even as load fluctuates



Engineering for predictability

Architectural improvements and new features don't affect performance

- Over 4 billion checksums per second for durability
- Strong consistency
- Encryption by default
- And many more

Scalability

Amazon S3 scales to virtually unlimited read or write throughput



Pay for
what you use

Amazon S3
balances
responsiveness
with efficiency

Engineering for scalability

Mostly, “it just works”

- Organic changes in load are detected and handled automatically
- Predictable performance

Some workloads benefit from a more proactive approach

- Published best practices help customers optimize performance

Using the scale of Amazon S3

Scaling through parallelization

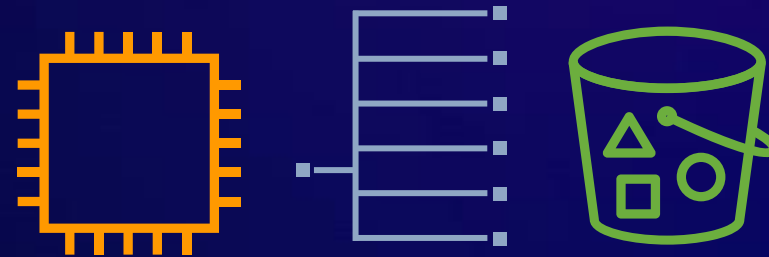
Concurrent requests



Fetch objects in parallel

One concurrent request for each 90–95 MB/s of desired network throughput

Parallel connections

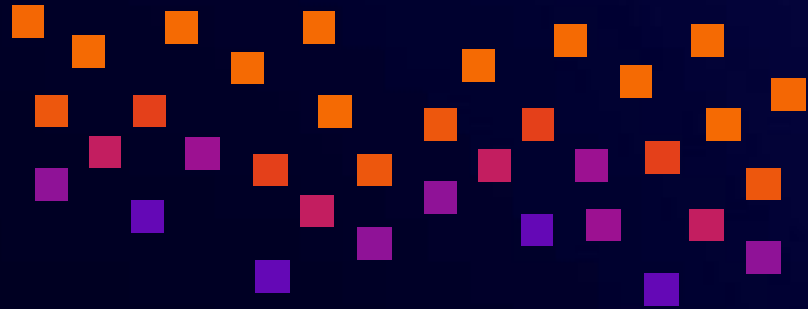


Connect with multiple Amazon S3 endpoints

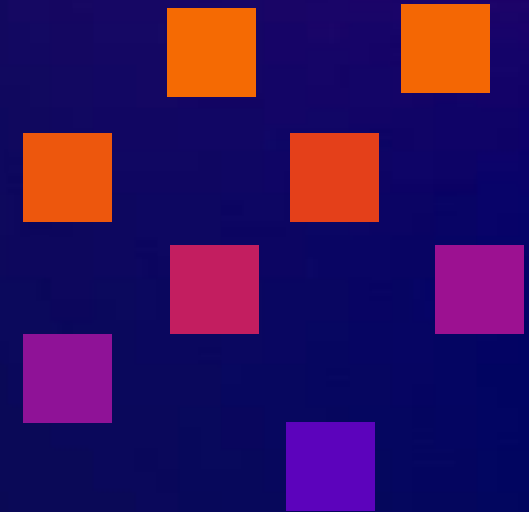
DNS lookups change every 5 s (TTL)

Use multivalued answers (up to 8) for alternative connection

Scaling dimensions



Transactions per second



Bytes per second

Performance engineering in the client

Best practices in the AWS Common Runtime (CRT) include

- - - Request parallelization
- - - Retry strategy with exponential backoff and jitter
- - - Connection reuse and management
- - - and more...

Mountpoint for Amazon S3 is built over the CRT

Scaling transactions per second

3,500

PUT, COPY, POST, DELETE



5,500

GET, HEAD

Per prefix



Amazon S3 Express One Zone

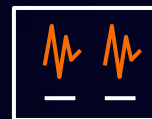
Highest performance object storage
in the cloud



Single-digit millisecond first byte
latency for compute-intensive and
latency-sensitive applications



Consistent performance eliminates tail
latencies, driving down query times



Data access speeds up to **10x faster**,
and request costs up to **50% lower**
than S3 Standard

Key takeaways

- Scale horizontally in both the bandwidth and request dimensions
- Parallel access to your datasets by connecting to multiple S3 endpoints
- Amortize overheads by reusing connections whenever possible
- AWS Common Runtime (CRT) is the easy button for client-side tuning
- For your most latency sensitive workloads, utilize S3 Express One Zone and get consistent single-digit latency access

Questions



Thank you!



Please complete the session
survey in the mobile app

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Not magic, just built differently

S3 EXPRESS ONE ZONE PERFORMANCE EXPLAINED



Introducing Directory Buckets

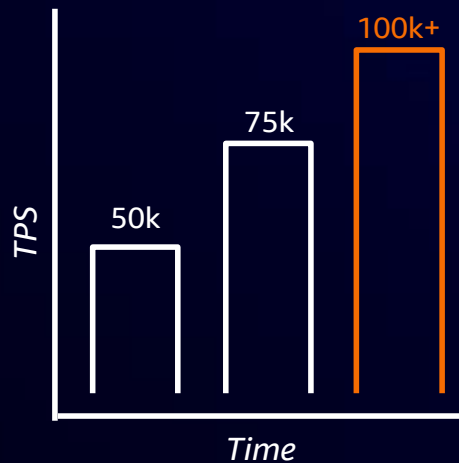
- One zone storage
- A new request scaling model
- Session-based authentication

Introducing Directory Buckets

A DIFFERENT REQUEST SCALING MODEL

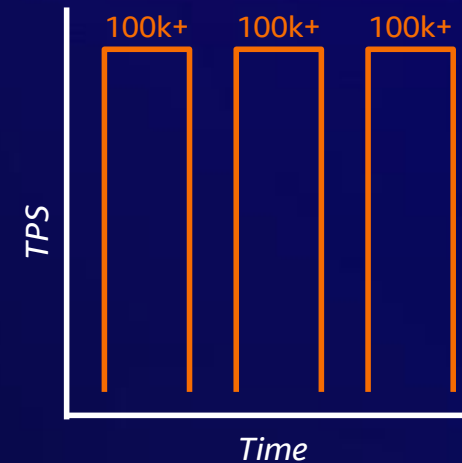
Regional Buckets

Scales by prefix,
adds TPS capacity under load



Directory Buckets

Scales by bucket,
all at once





Single-step Batch Operations

Easily move objects from regional
to directory buckets



Batch processing that scales to
thousands of objects per second, fully
managed within S3



Jobs can now process buckets and
prefixes, instead of relying on a
predefined manifest of objects



Purpose-built workflow in the S3
Console imports data sets into S3
Express with a single click



Automatic Caching for Mountpoint for Amazon S3

Reduce request cost and accelerate performance for repeat read requests



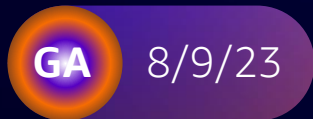
Files read from S3, then retained on EC2 Instance storage, EBS Volumes, or memory



First request at S3 latency, subsequent requests at local, sub-millisecond latency



Training jobs up to 2x faster by retaining data for repeated requests on Amazon EC2 instance storage

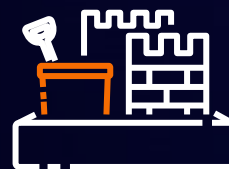


Mountpoint for Amazon S3 for Containers

Simplified file-based access to S3 from containers



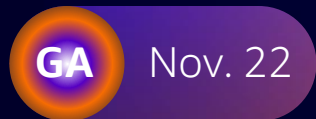
Simplified architecture through Automatically attached S3 storage to Kubernetes pods



Deploy Mountpoint as a CSI driver to your EKS cluster as an EKS add-on



Same Mountpoint performance and features, just containerized



Amazon S3 Connector for PyTorch

High-performance S3 primitives
for PyTorch training loops



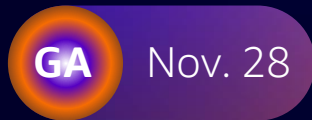
Delivers high throughput between
Amazon S3 and a PyTorch training job



Up to a 40% improvement to loading
times for both iterable-style and map-
style datasets



Up to a 40% improvement to
checkpoint performance by bypassing
local storage, streaming directly to S3



Mountpoint for Amazon S3 Express One Zone

High performance access to S3
Express One Zone objects



6x higher throughput for random
reads with S3 Express One Zone
relative to S3 Standard



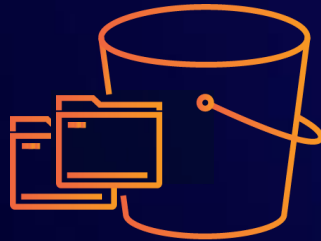
Accelerate machine learning training
jobs for vision models, finishing them
faster and saving on compute costs



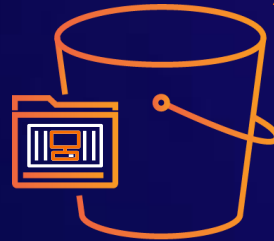
Training jobs using Mountpoint are up
to 30% faster with S3 Express One
Zone than with S3 Standard



Amazon S3
Mountpoint



Amazon S3
Mountpoint with
Local Cache



Amazon S3
Mountpoint for
Containers



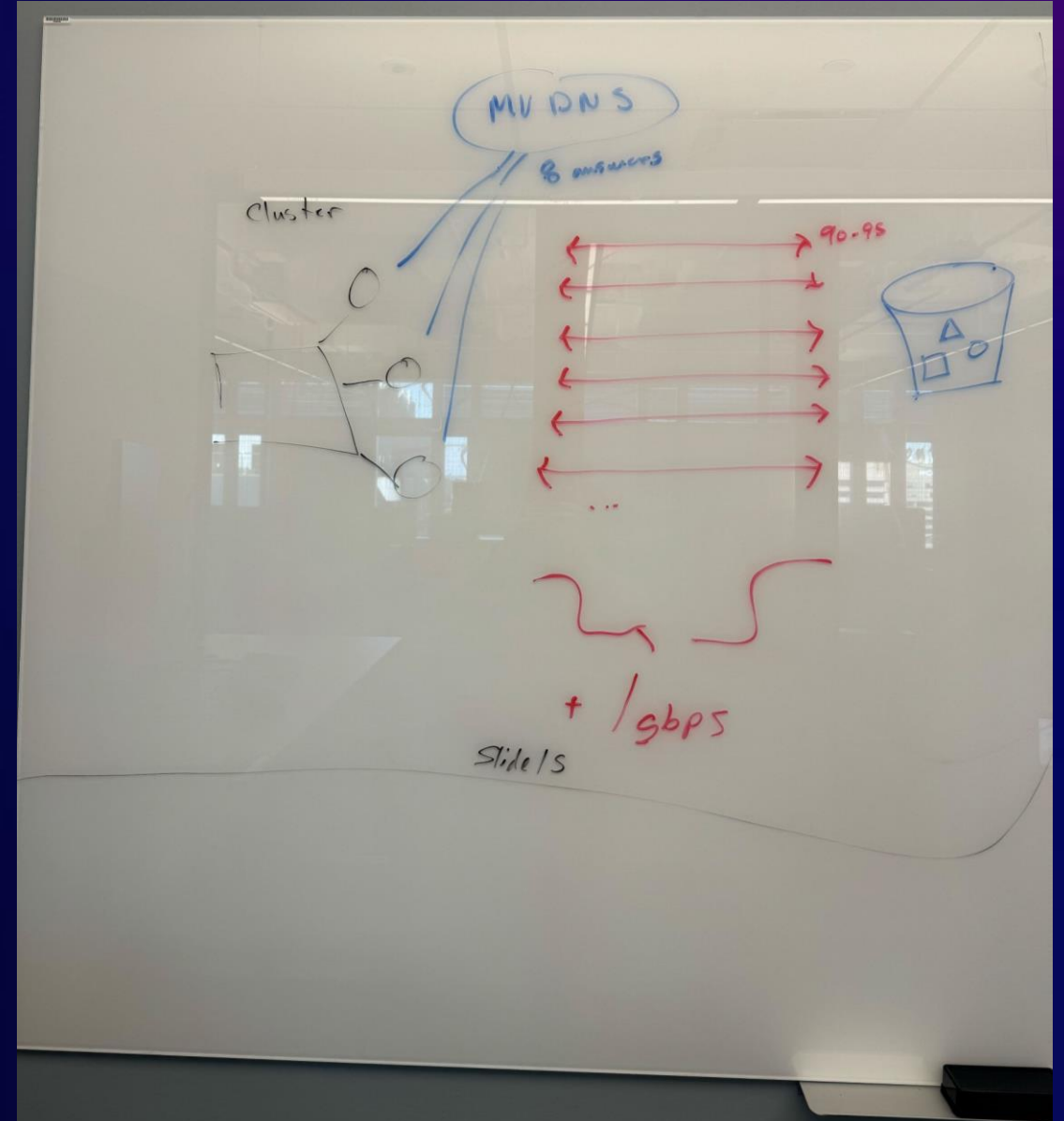
Amazon S3
Mountpoint for
S3 Express

The Amazon S3 Mountpoint Family

Chalkboard

Topic covered

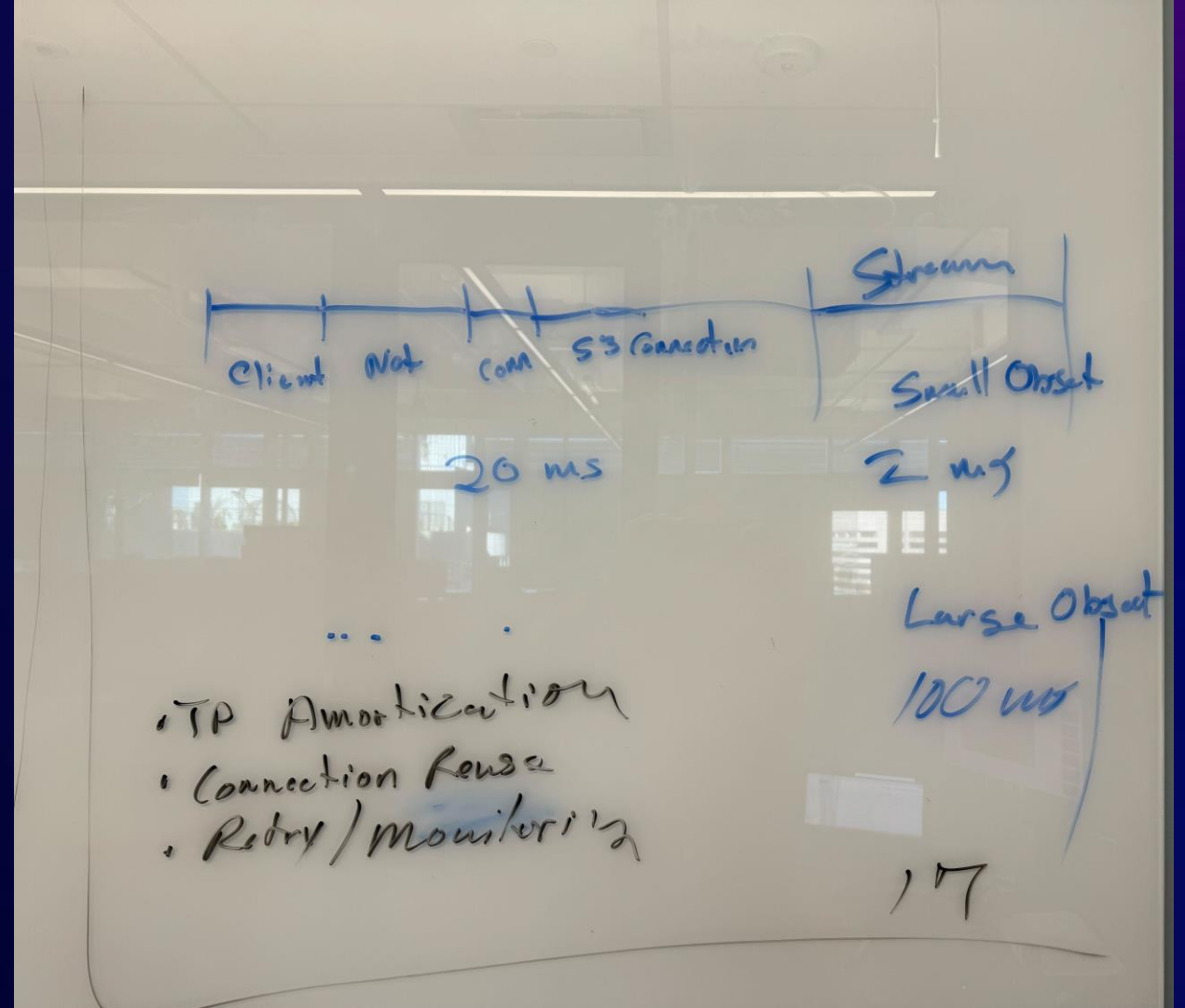
- Architecture diagram of client cluster accessing S3 bucket
- S3 endpoint provides MVA DNS responses (up to 8)
- Parallelizing connections
 - Writes – multipart upload
 - Reads – range reads
- Bandwidth expectation per connection – 90 to 95 Mbps



Chalkboard

Topic covered

- Object size matters
 - Larger object
 - Payload transfer dominates total operation latency
 - Initial connection establishment time is relatively smaller
 - Bytes per second
 - Parallelization by splitting up requests
 - Smaller object
 - Establishing connection dominates total operation latency
 - Transaction time is relatively large
 - Connection reuse
 - Retries



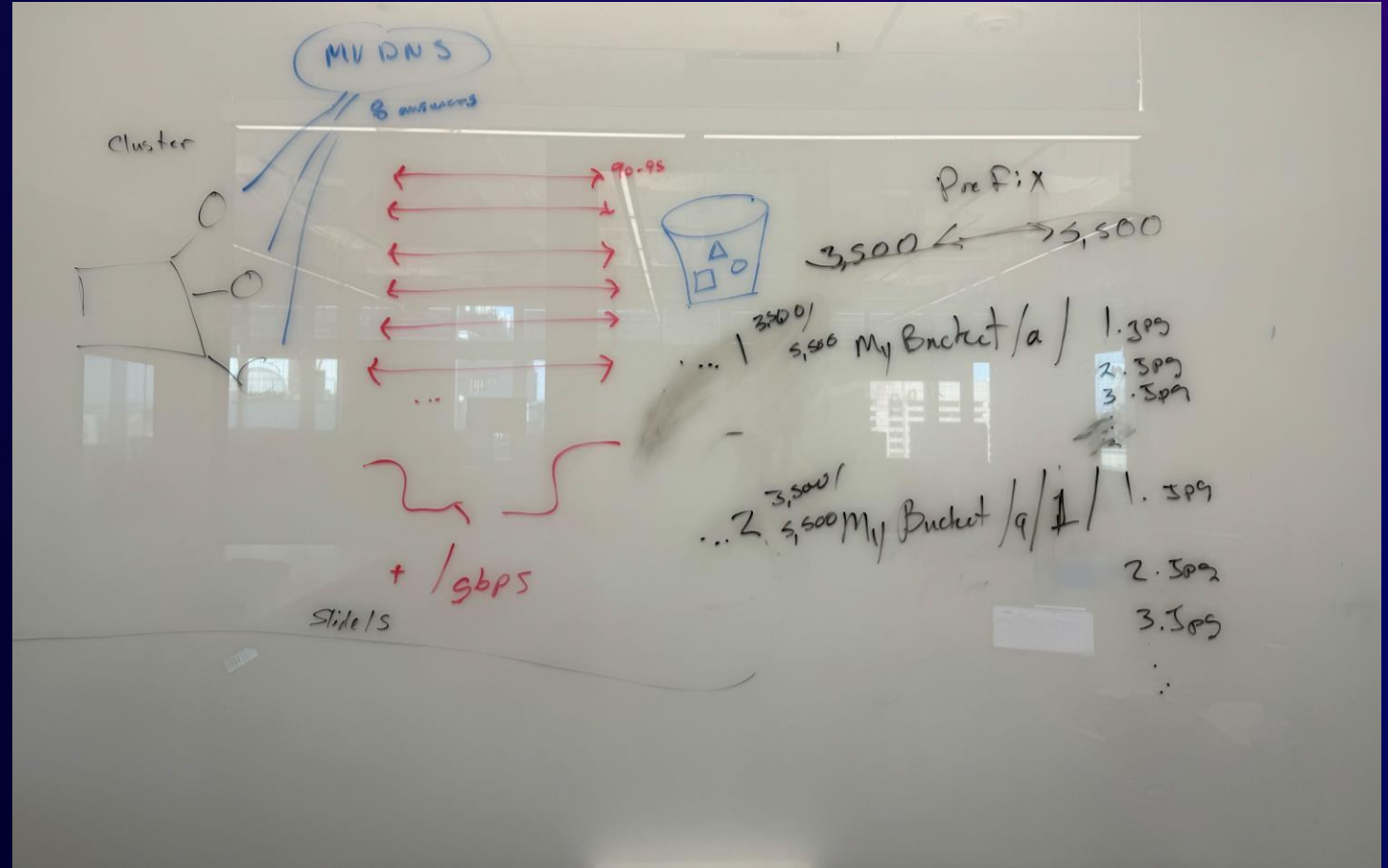
Chalkboard

Prefix design example

DOC-EXAMPLE-BUCKET/a/0001.jpg
DOC-EXAMPLE-BUCKET/a/0002.jpg
DOC-EXAMPLE-BUCKET/a/0003.jpg
...
DOC-EXAMPLE-BUCKET/a/9999.jpg



DOC-EXAMPLE-BUCKET/ax/0001.jpg
DOC-EXAMPLE-BUCKET/ax/0002.jpg
DOC-EXAMPLE-BUCKET/ax/0003.jpg
...
DOC-EXAMPLE-BUCKET/ax/9999.jpg



Chalkboard

Topic covered

- Sample use case for ML training with S3 Express Zonal
- Batch Operations Copy from S3 Regional to S3 Express Zonal
- Zonal – closer to compute (GPUs) – lower TTFB latency – sub-millisecond
- Auth model – reduces overhead for auth and connection establishment
- Latency ceiling – client retry faster
- Higher per-connection bandwidth
- Higher TPS with directory bucket – 100s of thousands of TPS

